Michael Huang

Mr. Allen

American Society

2026/4/24

The Impact of Atomic Bomb on the Expansion of American Imperialism

On August 7th, 1945, “First atomic bomb dropped on Japan; missile is equal to 20,000 tons of TNT; Truman warns foe a ‘rain of ruin’” (First Atomic Bomb) occupied the headlines of the New York Times. This news illustrated the first application of atomic bomb, the earliest nuclear weapon powered by nuclear fission in human history. On August 6th, 1945, the uranium bomb was dropped by the United States on Hiroshima, Japan, completely devastating 11.4 square kilometers in an instant. Though mass casualties resulted from this destructive force, it also provoked the ending of WWII, the greatest global conflict in human history, and made the United States the first nation to possess nuclear power. Meanwhile, this bombardment triggered the controversy regarding the influence of the atomic bomb on the expansion of American Imperialism. Although the atomic bomb is catastrophically destructive, it is still one of the main factors that shaped the world’s technology, geopolitical structure, and normative power, enabling America to establish a form of imperial dominance unprecedented in modern human history.

After the implementation of atomic bomb, the world was enveloped by the immense shock of this superpower, elevating the global status of the United States during the time. On the morning of August 6th, 1945, “Hiroshima was destroyed at one strike by a single atomic bomb dropped by a Superfort” (Nakashima). 70,000 people were killed, “the death toll is expected to reach 100,000, and people continue to die daily from burns suffered from the bomb's ultra-violet rays”

(Nakashima). The massive destruction in the city center and the increasing mortality instilled fear in Japanese leadership, prompting them to reconsider their war strategies, ultimately drawing to the surrender of Japanese imperialism.

On the same day, atomic bomb topped all news in London. *The Daily Express*'s headlined story "Atomic Bomb Warns Japs to surrender" declared that "the world has changed overnight", expressing the astonishment of the nuclear bomb's power and the relief of the imminent ending to the war. In the meantime, Winston Churchill, the prime minister of Great Britain, announced the Allies' success in the "grisly race between Germany and the Allies to find a weapon so destructive that it would ensure absolute victory in the world war" (Daniel). For the Soviet Union, however, American success in the nuclear bomb threatened its global status. As one of the two remaining superpowers, Moscow, the capital of the USSR, only focused its radio broadcasts on the scientific progress within the Soviet Union. Without mentioning any detail regarding the United States's accomplishments in the atomic bomb, this abnormal tranquility represented the collapse of the political equilibrium existing during World War II. Becoming the first nuclear power, the United States could dictate the postwar norms precisely. Therefore, the congratulation from London and the complete silence in Moscow were sufficient to prove that the postwar geopolitical playing field was tilting toward American power, further strengthening it globally after the night of August 6th, 1945.

While international responses highlighted the astounding power of the bomb and its political significance, they also underscored the extraordinary level of technological innovation achieved by the United States. Due to the Nazi dictatorship in Germany and the instability of European warfare, many Jewish scientists immigrated to the United States in search of safety and enriching research opportunities. Among these scholars were remarkable figures including J. Robert

Oppenheimer, Leo Szilard, Albert Einstein, and Enrico Fermi. These figures, having fled from the Fascist regimes in Europe, organized a project to exploit the fission process for military purposes, which later evolved into the Manhattan Project. "The discovery of nuclear fission was first communicated by Lise Meitner and Otto Frisch, two Jewish scientists who had fled Germany, to Niels Bohr in Copenhagen. Bohr later brought news of its potential to United States, where scientists like Albert Einstein recognized its military implications. Through Einstein's personal influence, the scientists persuaded the president Franklin D Roosevelt" (Levy and Scott-Clark), starting the research project with an initial budget of $6,000. All those scientists enhanced America's research capability, facilitating significant scientific research and progress.

Through atomic research, countless unprecedented technological innovations that extended beyond atomic bomb emerged. "The Manhattan project generated thousands of new inventions, as represented by patent claims processed in secret by the project, which if filed would have represented some 1% of all patents in force at the end of World War II" (Wellerstein). Through the Manhattan Project, the research on the atomic bomb required the development of entirely new industrial processes that demanded complex engineering solutions and vast infrastructure. Specifically, Uranium-235, the only isotope of fissionable uranium by neutrons of all energies, was successfully separated from Uranium-238; the first nuclear reactor was constructed to generate sufficient fissile material for the weapon; highly radioactive substance polonium was produced to be used as neutron sources in the bomb. These scientific accomplishments not only promoted innovation during war but also laid the path for future scientific research in America. Therefore, this technological monopoly, is not merely an academic achievement, it forged an innovation gap that no other nation could possibly surpass within decades, and the material foundation that sustained American imperial influence.

The aftermath of the atomic bomb did not just encourage further atomic research, but advanced medical applications and biology studies. In the book "Life Atomic: Radioisotopes in Science and Medicine", it documented breakthroughs that linked with atomic research, including cancer treatment using cobalt-60, medical diagnostic test, and cutting-edge understanding of human metabolic pathways. These enhancements were made possible by the availability of radioisotopes. After the World War II, the US government produced radioisotopes in some of the same nuclear reactors that had been built to produce material for nuclear weapons, which were sold to laboratories, hospitals and companies. The commercialization of radioisotopes allowed the United States to relate nuclear science with civilian fields, transforming the initially destructive power into the soft power, which pioneered the rest of the world.

From a political perspective, the geopolitical influence of the nuclear bomb was immediate and drastic. With the creation of nuclear bomb, America opened its era of nuclear monopoly, reshaping the world order. Before the WWII, global powers were more distributed, with the Soviet Union rising as a major power. However, after America demonstrated absolutely overwhelming nuclear superiority over USSR after the bombing of Hiroshima, the USSR was displaced from its dominant position and compelled to enter a desperate arms race to achieve parity. Replacing the Soviet Union, America became a dominant global power.

By possessing the atomic bomb as the "ultimate power", the United States dictated terms in postwar negotiations and established its global security umbrella, which was the indispensable standard that made American leadership irreplaceable. In the postwar agreement regarding the prevention of nuclear war between the United States and the Soviet Union, signed on June 22, 1973, "the United States and the Soviet Union agreed to make the removal of the danger of nuclear war and the use of nuclear weapons an 'objective of their policies,' to practice restraint in their relations toward each other and toward all countries, and to pursue a policy dedicated toward stability and peace"(US Department of State). Therefore, atomic bomb was no longer just a warfare weapon, it was used as a diplomatic tool to restrain Soviet Union's expansion in Eastern Europe and East Asia, forcing Stalin into more consistent peace terms. Moreover, on April 4th, 1949, America established the North Atlantic Treaty Organization, NATO, an organization involving 32 nations, providing a structure where European allies contribute significant military power under US. command. In the 1950s, "one of the first military doctrines of NATO emerged in the form of 'massive retaliation,' or the idea that if any member was attacked, the United States would respond with a large-scale nuclear attack" (Office of the Historian). Without the pioneering nuclear states, the United States had no unique offerings to the European allies; ultimately, with the nuclear superiority, American protection transformed the military alliance into an instrument for hegemonic expansion.

To constrain this destructive weapon, nuclear taboo emerged, which made the United States a vital component of the world dynamics. As depicted by the New York Times, "a single plane can carry an atomic bomb- a relatively small missile- that has an explosive force equivalent to 20,000 tons of TNT" (The Blasting Power). The two atomic bombs with a combined weight of 800 pounds were each dropped in Hiroshima and Nagasaki, killing 90,000 people, which amounted to 22,500 deaths per ton. Compared to the 3 deaths per ton of TNT during World War I in the bombing of London, the atomic bomb "slaughtered 75,000 times as many people per ton dropped in August 1945 as TNT bombs killed per ton in 1914 to 1918 in London" (Hart 277). This overwhelming increase in the destructive power of modern weapon precipitated the emergence of nuclear taboo, prohibiting nations from the first use of nuclear weapon even when deterrence mechanisms were not displayed. By delegitimizing the atomic bomb, the US government framed it as abhorrent, preserving the moral high ground of America. Condemning the moral aspects of the atomic bomb, America restrained nuclear proliferation, boycotting other countries from developing nuclear power. This rhetorical strategy defined the rules for modern warfare as America had already won under the prior rules. Therefore, the taboo not only applied to constrain nuclear use, but to legitimize American nuclear authority as an incomparable existence worldwide.

In short, the bombardment in Hiroshima on August 7th, 1945, marked the ending of World War II but also inaugurated a profound transition in global dynamics that accelerated the expansion of American Imperialism. The unprecedented destructive capacity of the atomic bomb propelled the United States toward superior military and political dominance, while reshaping international perceptions of power and deterrence. Beyond warfare, the technological and scientific advancements of the Manhattan Project reinforced America's long-term innovation capacity, permeating into the medical field, industrial development, and scientific research. Strategically, American nuclear monopoly stimulated the formation of global alliances, most notably NATO, enlarging the area covered by the American nuclear umbrella. Following the establishment of the nuclear taboo, the United States consolidated its authority and formulated the postwar international norms. Taken together, the atomic bomb undisputedly catalyzed American global dominance, strengthening its political, technological, and ideological leadership.

For the modern world, the contemporary political system consistently remained deeply influenced by the structure born in 1945. Nuclear deterrence, alliance systems, and nonproliferation regimes continue to govern the decisions made by every major power. Yet these frameworks, long presented as mechanisms upholding global stability, consistently reflect the strategic interests of the nation that built them. As the tensions between nuclear-armed countries intensify, the fragility of the system constructed by America becomes more conspicuous. The aggravation between China and America demonstrates the incremental erosion of the past global imperial system constructed by the US. Therefore, the atomic bomb is no longer just the weapon that ended World War II; rather, it engineered a world order that is still operating. Analyzing the effect of the atomic bomb in expanding American Imperialism, this research reveals the constructive legacy of the atomic bomb, including the world order it structured, the norms it established, and the national dominance of the United States is gradually institutionalized over time.

Reference

1. “Atomic Bombing of Japan Cost $2,000,000,000; Only 3 Other Projects Had Greater Top Priority.” *The New York Times*, 7 Aug. 1945, p. 5. *TimesMachine*, timesmachine.nytimes.com/timesmachine/1945/08/07/88273625.html.

2. Bernstein, Barton J. “The Atomic Bomb and American Foreign Policy, 1941–1945: An Historiographical Controversy.” *Peace and Change*, vol. 2, no. 1, 1974. *JSTOR*, www.jstor.org/stable/2538972.

3. "The Blasting Power of the New Bomb." *The New York Times*, 7 Aug. 1945, p. 3. *TimesMachine*, timesmachine.nytimes.com/timesmachine/1945/08/07/88273586.html?pageNumber=3. Accessed 29 Apr. 2026.

4. Daniel, Clifton. "Report by Britain: 'By God's Mercy' We Beat Nazis to Bomb, Churchill Says." *The New York Times*, 7 Aug. 1945, pp. 1, 9. *TimesMachine*, timesmachine.nytimes.com/timesmachine/1945/08/07/88273542.html?pageNumber=1. Accessed 29 Apr. 2026.

5. Freedman, Lawrence. “The Revolution in Strategic Affairs.” *Survival*, vol. 43, no. 2, 2001, pp. 7–26. https://doi.org/10.1080/00396330112331343155.

6. "First Atomic Bomb Dropped on Japan; Missile Is Equal to 20,000 Tons of TNT; Truman Warns Foe of a 'Rain of Ruin.'" *The New York Times*, 7 Aug. 1945, p. 1. *TimesMachine*, timesmachine.nytimes.com/timesmachine/1945/08/07/issue.html. Accessed 29 Apr. 2026.

7. Hart, Hornell. “Technological Acceleration and the Atomic Bomb.” *American Sociological Review*, vol. 11, no. 3, 1946, pp. 277–93. *JSTOR*, https://doi.org/10.2307/2087112.

8. Hotchkiss, Michael. “Beyond the Bomb: Atomic Research Changed Medicine, Biology.” *Princeton University*, 27 Feb. 2014, www.princeton.edu/news/2014/02/27/beyond-bomb-atomic-research-changed-medicine-biology.

9. “How the Press Reported the Atomic Bomb.” *American Experience*, PBS, www.pbs.org/wgbh/americanexperience/features/how-press-reported-atomic-bomb/. Accessed 17 Apr. 2024.

10. Levy, Adrian, and Catherine Scott-Clark. “Atomic Bomb: Development and Proliferation of Atomic Bombs.” *Encyclopædia Britannica*, www.britannica.com/technology/atomic-bomb/Development-and-proliferation-of-atomic-bombs. Accessed 17 Apr. 2024.

11. Nakashima, Leslie. "Hiroshima Gone, Newsman Finds." *The New York Times*, 31 Aug. 1945, p. 4. *TimesMachine*, timesmachine.nytimes.com/timesmachine/1945/08/31/88287746.html?pageNumber=4. Accessed 29 Apr. 2026.

12. “North Atlantic Treaty Organization (NATO), 1949.” *Office of the Historian, U.S. Department of State*, history.state.gov/milestones/1945-1952/nato. Accessed 28 Apr. 2026.

13. Tannenwald, Nina. “The Nuclear Taboo: The United States and the Normative Basis of Nuclear Non-Use.” *International Organization*, vol. 53, no. 3, 1999, pp. 433–68. *Cambridge Core*, doi:10.1162/002081899550957.

14. "Text of Statements by Truman, Stimson on Development of Atomic Bomb." *The New York Times*, 7 Aug. 1945, p. 3. *TimesMachine*, timesmachine.nytimes.com/timesmachine/1945/08/07/88273586.html.

15. "Treaty on the Non-Proliferation of Nuclear Weapons (NPT)." *U.S. Department of State*, 20 Jan. 2017, 2009-2017.state.gov/t/isn/5186.htm#narrative. Accessed 28 Apr. 2026.

16. "War Department Release on Atomic Bomb." *The New York Times*, 7 Aug. 1945, p. 4. *TimesMachine*, timesmachine.nytimes.com/timesmachine/1945/08/07/88273622.html.